

Tribonacci Numbers

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September 28, 2026

1 Definition

The Tribonacci sequence is defined as:

$$\begin{aligned}t_0 &= 0 \\t_1 &= 1 \\t_2 &= 1 \\t_n &= t_{n-1} + t_{n-2} + t_{n-3} \quad \forall \quad n \in \mathbb{Z}^+\end{aligned}\tag{1}$$

The first few Tribonacci numbers then would be:

$$0, 1, 1, 2, 4, 7, 13, 24, 44, 81, \dots$$

2 Ratios

Taking the ratios of consecutive Tribonacci numbers we find this pattern:

$$\begin{aligned}\frac{t_2}{t_1} &= 1 \\ \frac{t_3}{t_2} &= \frac{2}{1} \\ \frac{t_4}{t_3} &= \frac{2}{1} \\ \frac{t_5}{t_4} &= \frac{7}{4} \\ \frac{t_6}{t_5} &= \frac{13}{7} \\ \frac{t_7}{t_6} &= \frac{26}{13}\end{aligned}$$

Theorem 2.1. $\lim_{n \rightarrow \infty} \frac{t_{n+1}}{t_n} = \frac{1}{3} \cdot \left(1 + \sqrt[3]{19 + 3\sqrt{33}} + \sqrt[3]{19 - 3\sqrt{33}}\right)$
Proof TODO

3 Closed Form Expression

Using the characteristic equation of this sequence definition (see Characteristic Equation of recurrence relations), we have $x^3 - x^2 - x - 1 = 0$, with distinct roots of α, β, γ .

Theorem 3.1.

$$t_n = c_1 \cdot \alpha^n + c_2 \cdot \beta^n + c_3 \cdot \gamma^n \quad \forall n \in \mathbb{Z}^+$$

Where the constants c_1, c_2, c_3 are derived using the initial conditions.

Theorem 3.2.

$$t_n = \frac{\alpha^n}{-\alpha^2 + 4\alpha - 1} + \frac{\beta^n}{-\beta^2 + 4\beta - 1} + \frac{\gamma^n}{-\gamma^2 + 4\gamma - 1} \quad \forall n \in \mathbb{Z}^+$$

Proof. Using the initial conditions:

$$\begin{aligned} c_1 + c_2 + c_3 &= 0 & (n=0) \\ \alpha c_1 + \beta c_2 + \gamma c_3 &= 1 & (n=1) \\ \alpha^2 c_1 + \beta^2 c_2 + \gamma^2 c_3 &= 1 & (n=2) \end{aligned}$$

Rewriting this as a matrix equation gives a classic Vandermonde system:

$$\begin{bmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Using Gaussian elimination:

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ \alpha & \beta & \gamma & 1 \\ \alpha^2 & \beta^2 & \gamma^2 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & \beta - \alpha & \gamma - \alpha & 1 \\ 0 & \beta^2 - \alpha^2 & \gamma^2 - \alpha^2 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{\gamma - \alpha}{\beta - \alpha} & \frac{1}{\beta - \alpha} \\ 0 & 1 & \frac{\gamma^2 - \alpha^2}{\beta^2 - \alpha^2} & \frac{1}{\beta^2 - \alpha^2} \end{array} \right) \\ &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{\gamma - \alpha}{\beta - \alpha} & \frac{1}{\beta - \alpha} \\ 0 & 0 & \frac{\gamma^2 - \alpha^2}{\beta^2 - \alpha^2} - \frac{\gamma - \alpha}{\beta - \alpha} & \frac{1}{\beta^2 - \alpha^2} - \frac{1}{\beta - \alpha} \end{array} \right) \end{aligned}$$

Therefore we can solve for c_3 :

$$\begin{aligned} c_3 &= \frac{\frac{1}{\beta^2 - \alpha^2} - \frac{1}{\beta - \alpha}}{\frac{\gamma^2 - \alpha^2}{\beta^2 - \alpha^2} - \frac{\gamma - \alpha}{\beta - \alpha}} \\ &= \frac{(\beta - \alpha) - (\beta^2 - \alpha^2)}{(\gamma^2 - \alpha^2)(\beta - \alpha) - (\gamma - \alpha)(\beta^2 - \alpha^2)} \\ &= \frac{\beta - \alpha}{(\beta - \alpha)(\gamma - \alpha)} \cdot \frac{1 - (\beta + \alpha)}{(\gamma + \alpha) - (\beta + \alpha)} \\ &= \frac{1 - (\beta + \alpha)}{(\gamma - \beta)(\gamma - \alpha)} \\ &= \frac{1 - (\beta + \alpha + \gamma) + \gamma}{(\gamma - \beta)(\gamma - \alpha)} \end{aligned}$$

Since α, β and γ satisfy $x^3 - x^2 - x - 1 = 0$, Vieta's formulas give:

$$\alpha + \beta + \gamma = 1, \quad \alpha\beta\gamma = 1$$

Using these identities:

$$\begin{aligned} c_3 &= \frac{\gamma}{\gamma^2 - (\alpha + \beta)\gamma + \alpha\beta} \\ &= \frac{\gamma}{\gamma^2 - (\alpha + \beta + \gamma)\gamma + \gamma^2 + \frac{\alpha\beta\gamma}{\gamma}} \\ &= \frac{\gamma}{2\gamma^2 - \gamma + \frac{1}{\gamma}} \\ &= \frac{1}{2\gamma - 1 + \frac{1}{\gamma^2}} \end{aligned}$$

Rewriting $\frac{1}{\gamma^2}$ using $\gamma^3 - \gamma^2 - \gamma - 1 = 0 \implies \frac{1}{\gamma^2} = 2\gamma - \gamma^2$:

$$c_3 = \frac{1}{-\gamma^2 + 4\gamma - 1}$$

By symmetry:

$$c_1 = \frac{1}{-\alpha^2 + 4\alpha - 1}, \quad c_2 = \frac{1}{-\beta^2 + 4\beta - 1}$$

Hence:

$$t_n = \frac{\alpha^n}{-\alpha^2 + 4\alpha - 1} + \frac{\beta^n}{-\beta^2 + 4\beta - 1} + \frac{\gamma^n}{-\gamma^2 + 4\gamma - 1} \quad \forall n \in \mathbb{Z}^+$$

□

4 Negative Indices

If we ignore the condition of $n > 0$ we can use the functional form definition of Tribonacci numbers to define negative indices:

$$t_{n-3} = t_n - t_{n-1} - t_{n-2} \quad (2)$$

Therefore the first few will be:

$$\begin{aligned} t_{-1} &= t_2 - t_1 - t_0 = 1 - 1 - 0 = 0 \\ t_{-2} &= t_1 - t_0 - t_{-1} = 1 - 0 - 0 = 1 \\ t_{-3} &= t_0 - t_{-1} - t_{-2} = 0 - 0 - 1 = -1 \\ t_{-4} &= t_{-1} - t_{-2} - t_{-3} = 0 - 1 - (-1) = 0 \\ t_{-5} &= t_{-2} - t_{-3} - t_{-4} = 1 - (-1) - 0 = 2 \\ t_{-6} &= t_{-3} - t_{-4} - t_{-5} = -1 - 0 - 2 = -3 \end{aligned}$$

Theorem 4.1.

$$t_{-n} = t_{n-1}^2 - t_{n-2} \cdot t_n$$

Proof by induction. Assume true for $k, k+1$, and $k+2$:

$$\begin{aligned} t_{-k} &= t_{k-1}^2 - t_{k-2} \cdot t_k \\ t_{-k-1} &= t_{k-2}^2 - t_{k-3} \cdot t_{k-1} \\ t_{-k-2} &= t_{k-3}^2 - t_{k-4} \cdot t_{k-2} \end{aligned}$$

Using the linear recurrence $t_{-k-3} = t_{-k} - t_{-k-1} - t_{-k-2}$, algebraic substitution confirms the relation for $k+3$. $\square$